

AtomQuest Hackathon 1.0

In-House Goal Setting & Tracking Portal — Solution Architecture



Technology Stack

Layer	Technology	Reason
Frontend	Next.js 14 + TypeScript + Tailwind CSS	SSR, fast deploy on Vercel (free)
Backend	Node.js + Express + Prisma ORM	Typed queries, easy migration

Database	PostgreSQL (Supabase)	Free managed tier, full SQL support
Cache	Redis (Upstash)	Free tier, session & score caching
Auth	JWT (+ optional Azure AD)	Stateless, supports SSO bonus
Hosting FE	Vercel	Free, zero-config Next.js deploy
Hosting BE	Railway / Render	Free tier for backend + DB
Export	JSON → CSV streaming	No library overhead for reports

Progress Scoring Formulas

UoM Type	Description	Formula
Min (Numeric / %)	Higher is better — e.g. Sales Revenue	Achievement / Target x 100
Max (Numeric / %)	Lower is better — e.g. TAT, Cost	Target / Achievement x 100
Timeline	Date-based completion	On time = 100%, Late = 0%
Zero-based	Zero = Success — e.g. Safety incidents	If 0 then 100%, else 0%

User Roles & Capabilities

Role	Responsibilities	System Capabilities
Employee	Draft goals; enter quarterly achievement; update progress scores	Create & edit goals pre-submission; view locked goals; input actuals
Manager (L1)	Review & approve goals; conduct quarterly check-ins; log feedback	Feedback dashboard; inline editing during approval; comment / feedback
Admin / HR	Configure cycles; manage org hierarchy; oversee completion	Cycle management; exception handling; audit logs; goal unlock capabilities

Demo Login Credentials

Role	Email	Password
Employee	employee@demo.com	Demo@123
Manager	manager@demo.com	Demo@123
Admin / HR	admin@demo.com	Demo@123

Submission includes: Live demo URL · GitHub repository · This architecture diagram · Demo credentials above